Toward a Mechanistic DEB Model for *Aporrectodea caliginosa* - Linking Food Availability, Organic Matter Quality, and Energy Assimilation

Supplementary information

Table of contents

List of Figures

List of Tables

1 Earthworm culture

1.1 Breeding and experiment soil

The soil used in the culture of earthworms is collected from the top 10 cm of a meadow in Versailles called "Les Closeaux". It is then dried before being milled at < 2 mm.

Table 1.1: Main characteristics of the breeding soil from Versailles. Table from Bart et al. (2017).

Characteristics	Value
Clay (< 2 µm, g/kg)	226.0
Fine silt (2-20 µm, g/kg)	174.1
Coarse silt (20-50 µm, g/kg)	298.9
Fine sand (50-200 µm, g/kg)	239.1
Coarse sand (200-2000 µm, g/kg)	47.9
$CaCO_3$ total (g/kg)	23.3
Organic matter (g/kg)	32.6
P_2O_5 (g/kg)	0.1
Organic carbon (g/kg)	18.9
Total nitrogen (N) (g/kg)	1.5
C/N	12.5
pH	7.5
Total Cu (mg/kg)	25.2

1.2 DNA barcoding

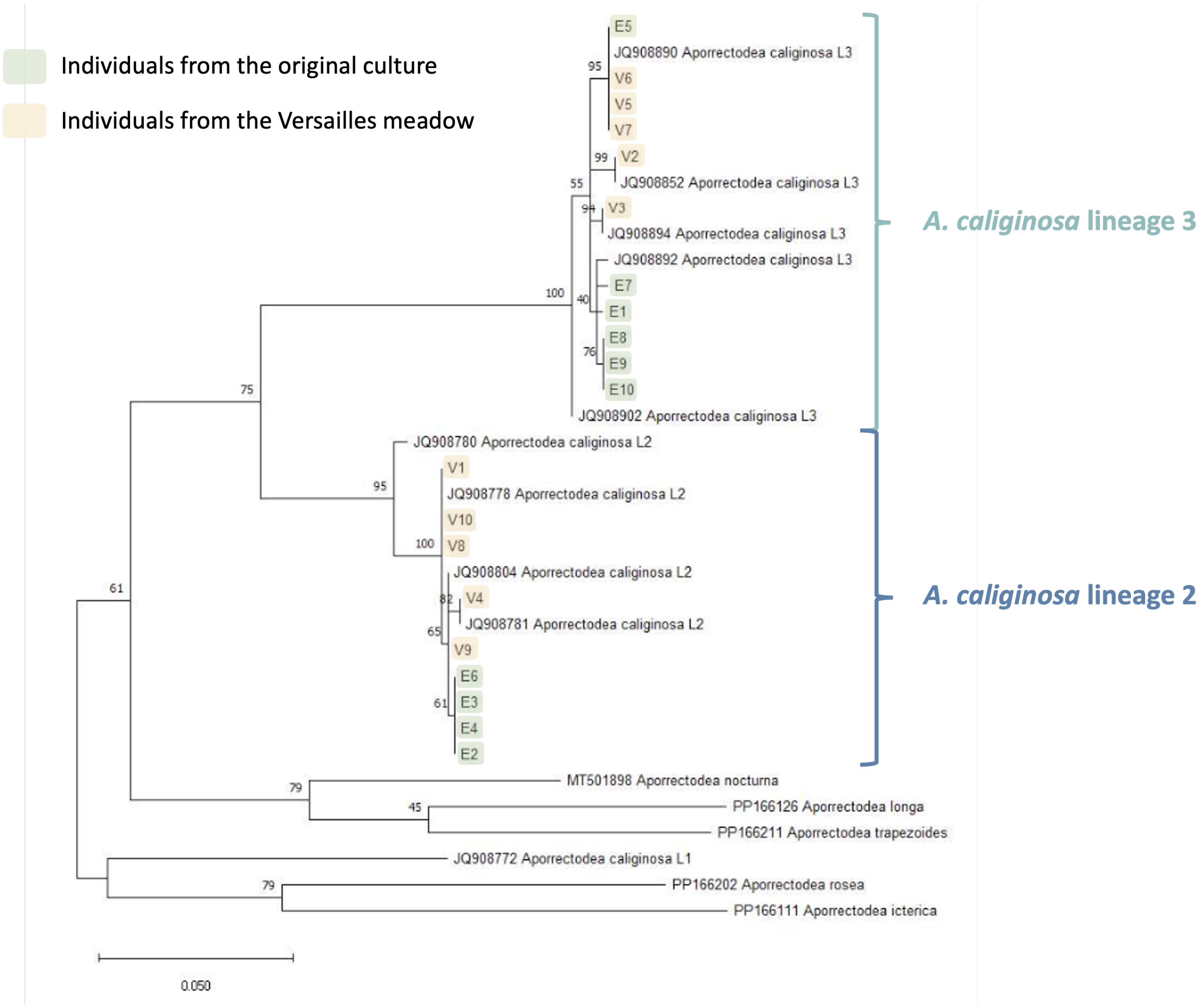


Figure 1.1: Results of the DNA barcoding of 10 individuals from the culture (highlighted in yellow)

The phylogenetic tree was realised thanks to maximum likelyhood bootstrap method (1000 simulations) with the Tamura-Nei model (nucleotides). Bootstrap frequency is indicated on each branch.

Aporrectodea caliginosa is a cryptic species complex comprising three lineages. Our earthworm culture is known to include individuals from two closely related lineages (2 and 3). While these lineages could potentially exhibit differences in sensitivity, previous research on *Gammarus roeselii* has shown that tolerance within cryptic species complexes is not always lineage dependent (Kabus et al., 2024).

2 Experiments

The Table 2.1 presents the characteristics of each experiment used to calibrate the DEB model.

Table 2.1: Summary of the experimental conditions and measured endpoints for each experiment used to calibrate the DEB model. The numbering for Bart (2019) follows that of the original publication. Control individuals from Bart et al. (2020) include those from the XP2 experiment reported in Bart (2019).

Reference	Experiment name	Soil in cosm	Food (g/ind/14d)	Density (#/cosm)	Temperature (°C)
Bart (2019)	XP1_1	Closeaux soil (400 g, WHC 70%)	3	1	15
	X1_2	Closeaux soil (400 g, WHC 70%)	3	1 then 2 from day 112, 3 from day 210 and 6 from day 294	15
	XP_3	Closeaux soil (400 g, WHC 70%)	0 but 3 at days 70, 84 and 140	1	15
	XP4_1	Closeaux soil (400 g, WHC 70%)	1	1	15
	XP4_2	Closeaux soil (400 g, WHC 70%)	1	1 then 2 from day 140, 3 from day 224 and 6 from day 322	15
	XP5	Closeaux soil (400 g, WHC 70%)	1.5	1	15
Bart et al. (2020)	Control individuals	Closeaux soil (400 g, WHC 70%)	3	1	15

Reference	Experiment name	Soil in cosm	Food (g/ind/14d)	Density (#/cosm)	Temperature (°C)
Present paper	D1	Closeaux soil (200 g, WHC 70%)	3	1	18
	D2	Closeaux soil (200 g, WHC 70%)	1.5	2	18
	D2b	Closeaux soil (200 g, WHC 70%)	3	2	18
	D5	Closeaux soil (200 g, WHC 70%)	0.6	5	18
	D7	Closeaux soil (200 g, WHC 70%)	0.4	7	18

3 Models

3.1 Model formulation

3.1.1 Model state and intermediate variables

Table 3.1: Model state and intermediate variables.

Symbol	Description	Unit
Environmental state variables		
OM_{horse}	Horse dung organic matter mass	g_{OM}
OM_{soil}	Soil organic matter mass	g_{OM}
Organism state variables		
E	Energy reserve	J
E_h	Maturity	J
L	Volumetric structural length	cm
R	Cumulated reproduction	#
W_w	Wet weight	g
Intermediate variables		
E_m	$\frac{\{\dot{p}_{Am}\}}{v}$	$J.cm^{-1}$
f	Scaled functional response	$-$
g	$\frac{[E_G]}{\kappa_{real} E_m}$	cm^{-2}
K_m	$\frac{[\dot{p}_M]_t}{[E_G]}$	d^{-1}
κ_{real}	Allocation fraction to soma	$-$
L_m	$\frac{v_t}{K_m g}$	cm
$\dot{OM}_{out}^{horse}$	Rate of consumption of horse dung organic matter	$g_{OM}.d^{-1}$
$\dot{OM}_{out}^{soil}$	Rate of consumption of soil organic matter	$g_{OM}.d^{-1}$
$\dot{p}_C$	Flux of mobilized energy from reserve	$J.d^{-1}$

Symbol	Description	Unit
$\dot{p}_R$	Energy flux allocated to maturity / reproduction	J.d^{-1}
$\dot{Q}_{f,max}$	Maximum rate of energy ingested	J.d^{-1}
$\dot{Q}_f^{horse}$	Rate of energy ingested through horse dung	J.d^{-1}

3.1.2 Standard DEB model

Reserve energy dynamic

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{\{\dot{p}_{Am}\}_t}{L} \left(f - \frac{E\ \nu_t}{\{\dot{p}_{Am}\}_t} \right)$$

with :

- E : Energy in the reserve (J)
- ν_t : Energy conductance at temperature T_{exp} $(cm.d^{-1})$

Growth

$$\frac{\mathrm{d}L}{\mathrm{d}t} = \frac{\nu_t}{3\ (\frac{E}{E_m} + g)} \left(\frac{E}{E_m} - \frac{L}{L_m} \right)$$

$$K_m = \frac{[\dot{p}_{M,t}]}{[E_G]}$$
$$E_m = \frac{\{\dot{p}_{Am}\}_t}{\nu_t}$$
$$g = \frac{[E_G]}{\kappa\ E_m}$$
$$L_m = \frac{\nu_t}{K_m\ g}$$

with :

- L : Structural volumetric length (cm)
- $[\dot{p}_{M,t}]$: Volume-specific maintenance rate $(J.d^{-1}.cm^{-3})$
- $[E_G]$: Volume-specific cost for structure $(J.cm^{-3})$
- κ : Energy allocation fraction to soma (-)

$$W_W = L^3 \left(1 + \frac{E}{E_m} w\right)$$

with :

- W_W : Wet weight of the earthworm (g)
- w : Contribution of reserve to body weight ($g.cm^{-3}$)

Maturity and reproduction

$$\dot{p}_C = \frac{g\ E}{g + \frac{E}{E_m}} (\nu_t\ L^2 + K_m\ L^3)$$

with :

- $\dot{p}_C$: Mobilized energy flux from reserve ($J.d^{-1}$)

$$\dot{p}_R = (1 - \kappa)\ \dot{p}_C - \dot{k}_{J,t}\ E_h$$

with :

- $\dot{p}_R$: Allocated energy flux to reproduction ($J.d^{-1}$)
- $\dot{k}_{J,t}$: Maturity maintenance rate (d^{-1})

$$\frac{\mathrm{d}E_H}{\mathrm{d}t} = \begin{cases} E_H \leq E_H^p, & \dot{p}_R \\ E_H > E_H^p, & 0 \end{cases}$$

with :

- E_h : Maturity (J)
- E_h^p : Maturity at puberty (J)

$$\frac{\mathrm{d}R}{\mathrm{d}t} = \begin{cases} E_H \leq E_H^p, & 0 \\ E_H > E_H^p, & \kappa_R\ \frac{\dot{p}_R}{E_0} \end{cases}$$

with :

- R : Cumulated reproduction (#)
- κ_R : Reproduction efficiency (-)
- E_0 : Energy cost for an egg (J)

Effect of temperature on rates

$$
\begin{aligned}
s_A &= \exp(\frac{T_A}{T_{ref} - \frac{T_A}{T_{exp}+273.15}})\\
s_{rH} &= \frac{1+\exp(\frac{T_A^H}{T^H} - \frac{T_A^H}{T_{ref}+273.15})}{1+\exp(\frac{T_A^H}{T^H} - \frac{T_A^H}{T_{exp}+273.15})}\\
T_c &= s_A(T_{exp}+273.15 \geq T_{ref})s_{rH} + (T_{exp}+273.15 < T_{ref})
\end{aligned}
$$

$$
\begin{aligned}
\{\dot{p}_{Am}\}_t &= \{\dot{p}_{Am}\}\ T_c\\
[\dot{p}_{M,t}] &= [\dot{p}_M]\ T_c\\
\nu_t &= \nu\ T_c\\
\dot{k}_{J,t} &= \dot{k}_J\ T_c
\end{aligned}
$$

with :

- T_A : Arrhenius temperature (K)
- T_A^H : Arrhenius temperature for upper boundary (K)
- T^H : Upper boundary (K)
- T_{ref} : Reference temperature (K)
- T_{exp} : Experimental temperature (°C)

3.2 Priors used

Priors are the same between models :

- Distrib (Fm, TruncNormal, 0.1, 0.2, 0.0001, 0.8);
- Distrib (pAm, TruncNormal_cv, 100, 0.2, 40, 2000);
- Distrib (r_pAm_pM, TruncNormal_cv, 0.8, 0.1, 0.1, 2);
- Distrib (v, TruncNormal_cv, 0.18505, 0.1, 0.001, 2);
- Distrib (kap, Uniform, 0, 1);
- Distrib (kapJ, Uniform, 0, 1);
- Distrib (kapA, Uniform, 0, 1);
- Distrib (Ehp, TruncNormal_cv, 100, 0.2, 20, 300);
- Distrib (E_coc, TruncNormal_cv, 30, 0.2, 1, 300);
- Distrib (rOM_ClxHorse, LogUniform, 0.00001, 0.3);
- Distrib (K, Uniform, 0.0001, 1);
- Distrib (dv, TruncNormal_cv, 1, 0.05, 0, 4);
- Distrib (wE, LogUniform, 1, 1000);
- Distrib (de, LogUniform, 0.000000001, 0.1);
- Distrib (tau, TruncNormal_cv, 7, 0.5, 0, 50);

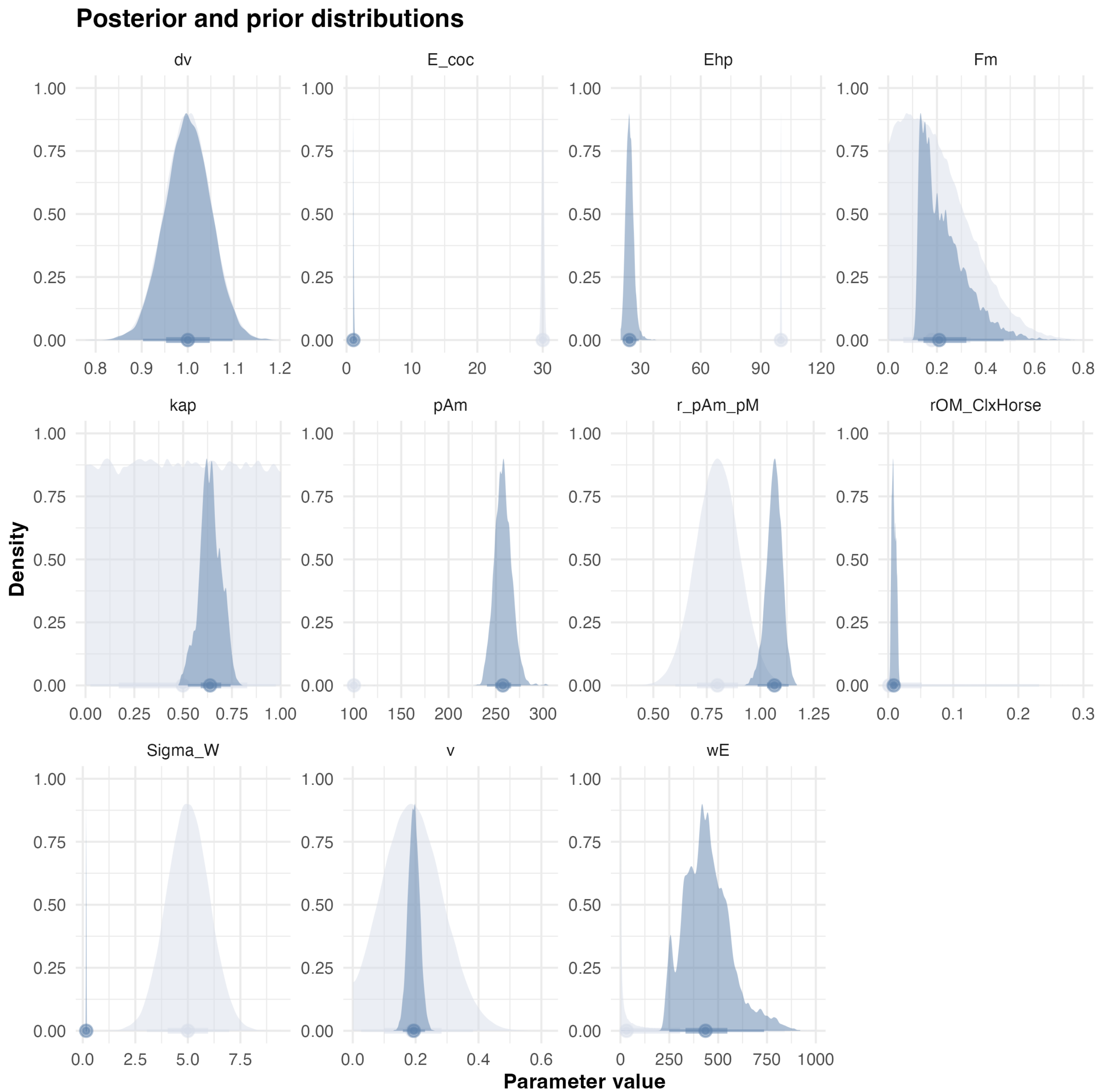
Log-Likelihood calculations :

- Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);
- Likelihood (Weight, Normal, Prediction(Weight), Sigma_W);
- Likelihood (Reproduction, Poisson, Prediction(Reproduction));

For the maximum ingestion rate, we imposed a biologically realistic constraint based on the assumption that an earthworm cannot ingest more than its own body weight per day. This constraint was used to bound the parameter $[\dot{F}_m]$.

An estimate of cocoon energy content was required to parameterize reproduction in the DEB model for *Aporrectodea caliginosa*. As no direct measurements are available for this species, we relied on published calorimetric data for *Lumbricidae* reported by Cumminns & Wuycheck (1971), who measured the energy density of whole individuals across developmental stages. Reported mean energy densities were 5,690 cal.g^{-1} dry weight ($n = 3$), corresponding to 19,098 J.g^{-1}, and 782 cal.g^{-1} wet weight ($n = 1$), corresponding to 3,268 J.g^{-1}. Although these measurements do not specifically target cocoons, they provide a reasonable order-of-magnitude estimate for annelid biomass. Assuming a mean cocoon fresh mass of 10 mg, this yields an approximate energy content of 33 J per cocoon. Comparable values have been reported for soil invertebrates in other calorimetric studies (e.g., Driver et al. (1974); Eggleton & Schramm (2004)), supporting the use of this approximation as a prior for egg energy content.

3.3 Model A_FLYw



{fig-scap="Estimated posterior distributions for each parameter."}

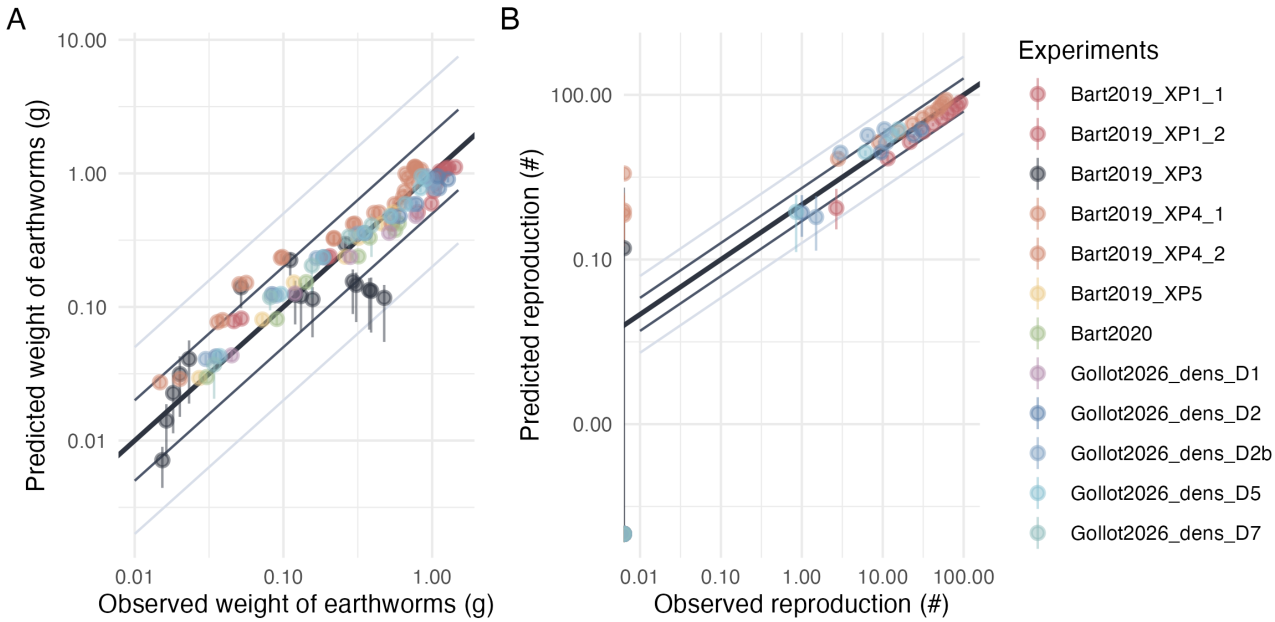


Figure 3.1: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

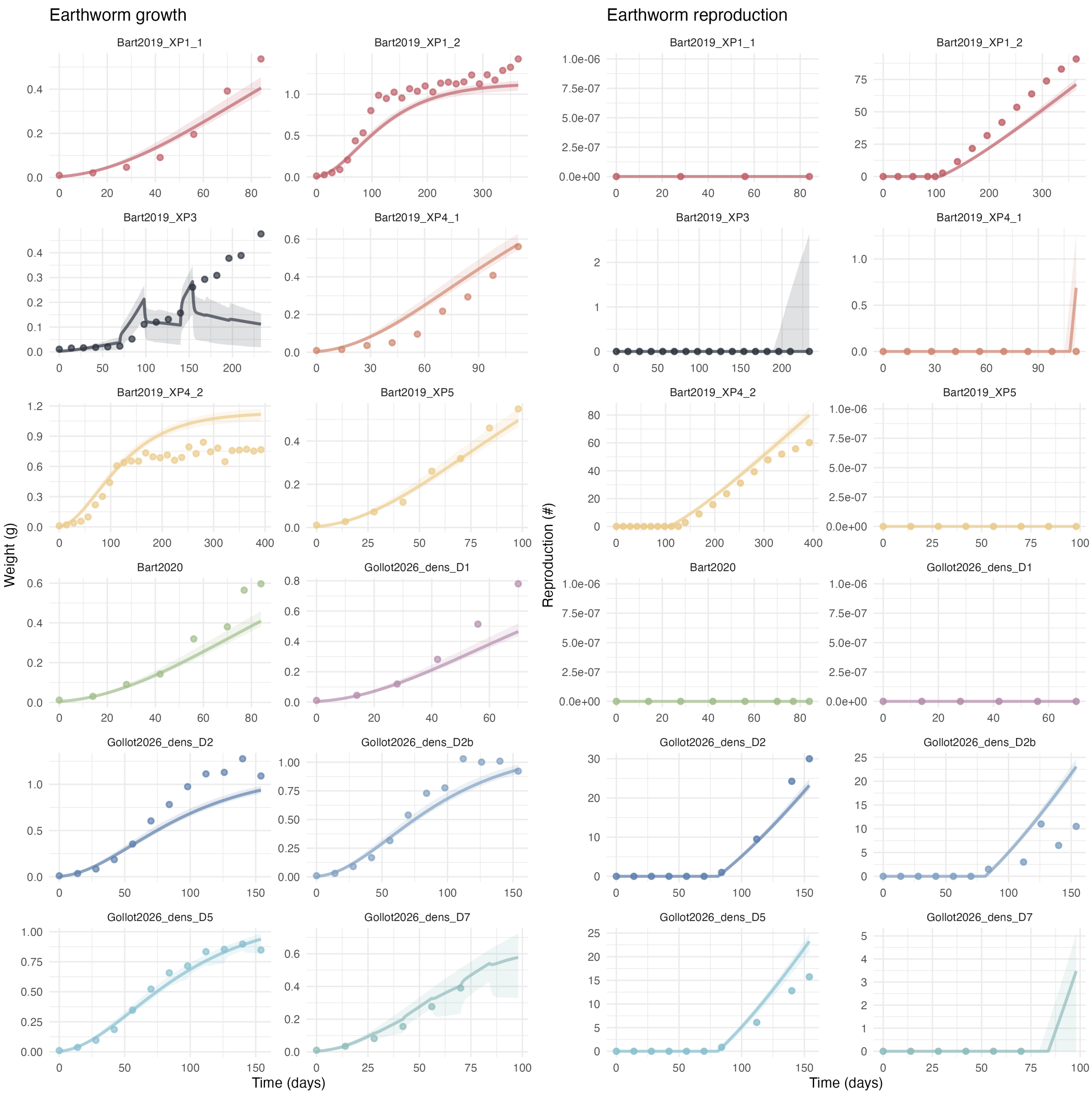


Figure 3.2: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.4 Model B_FLYe

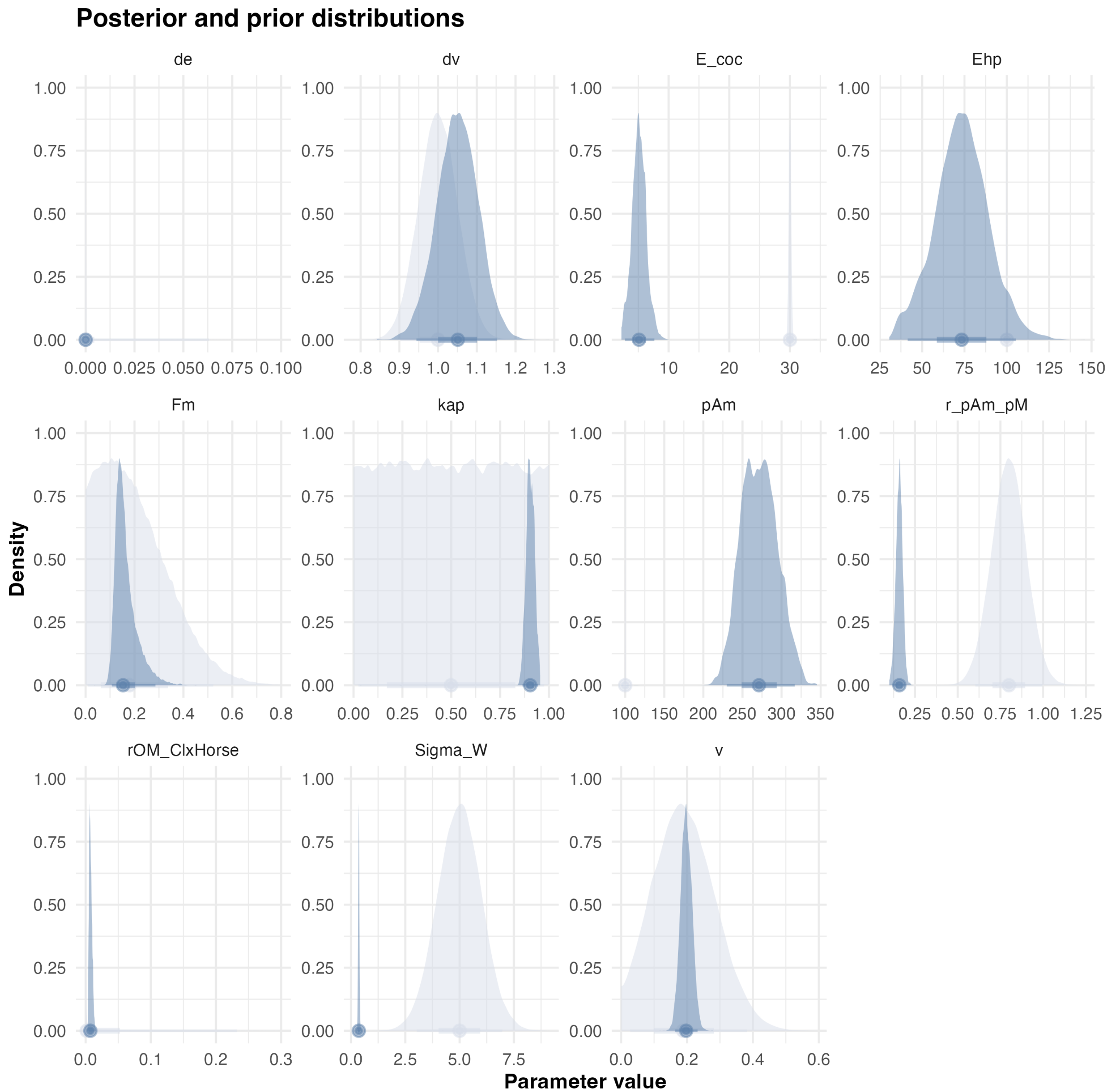


Figure 3.3: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

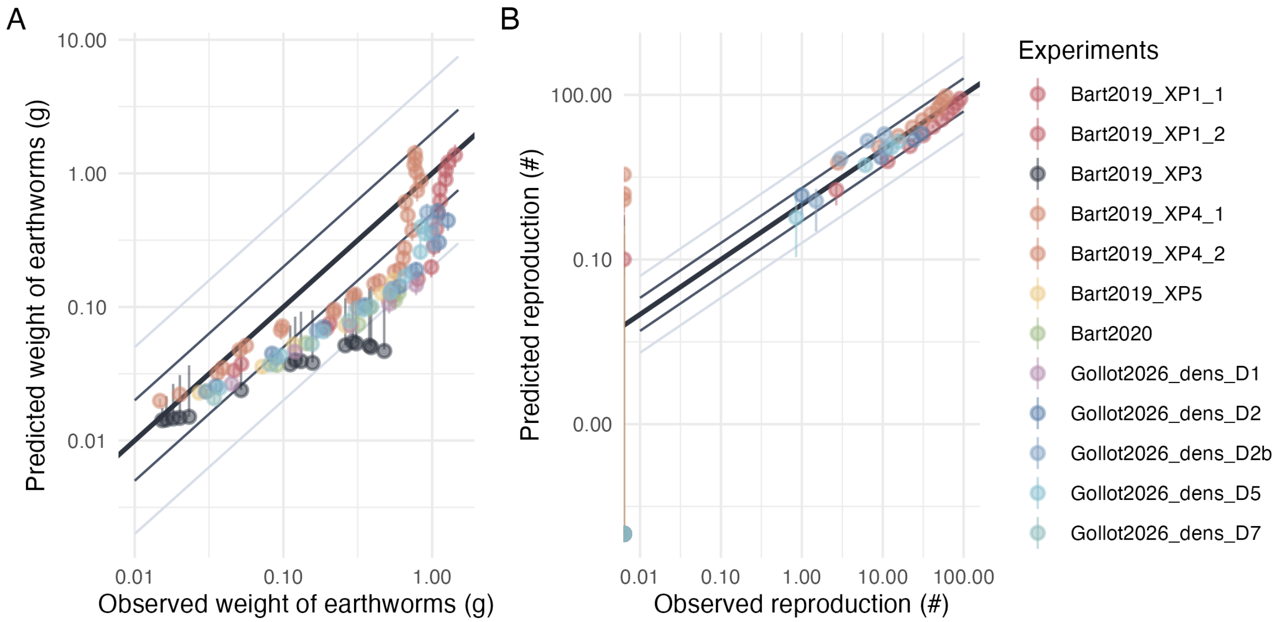


Figure 3.4: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

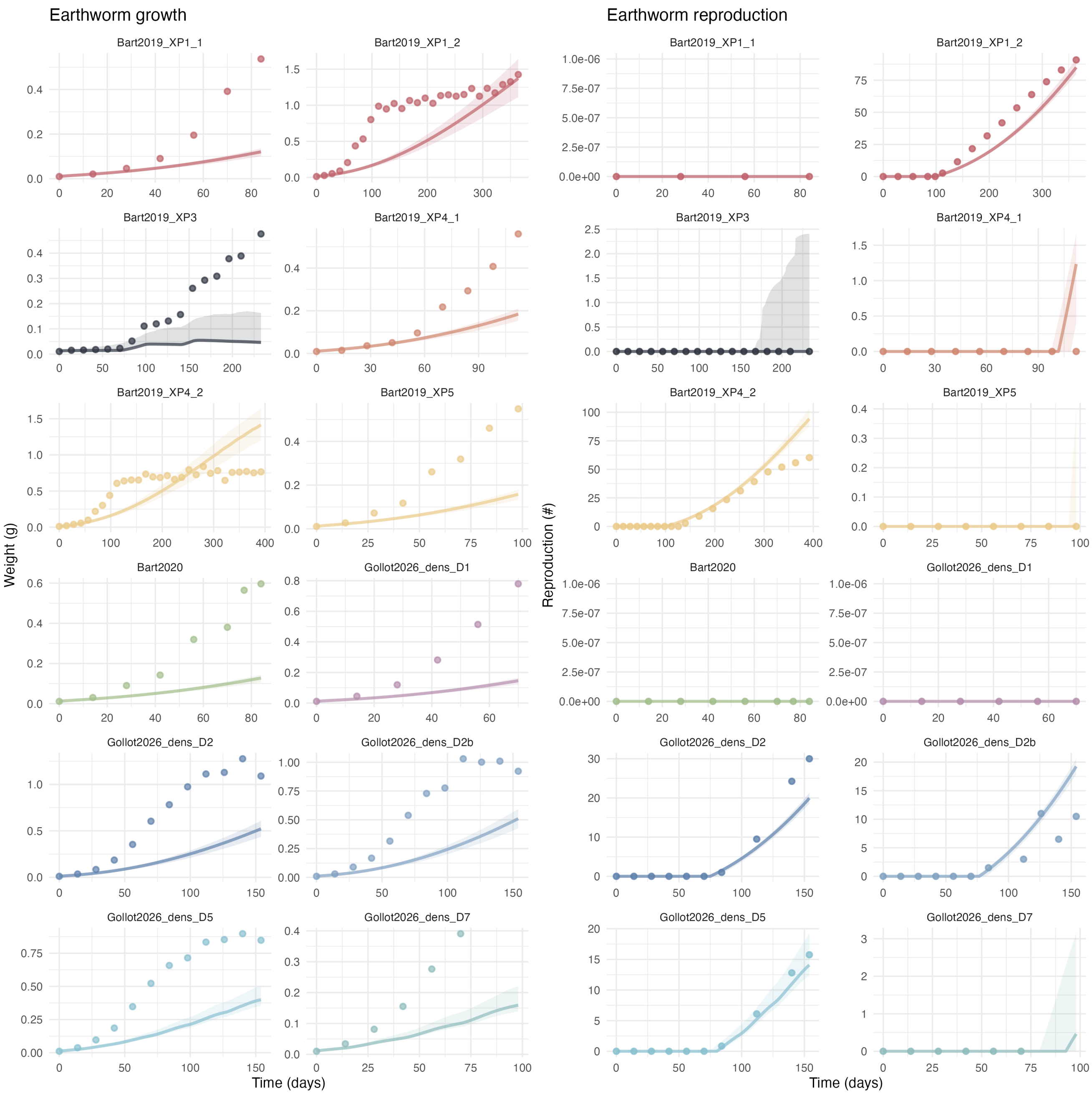


Figure 3.5: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.5 Model C_CWYe

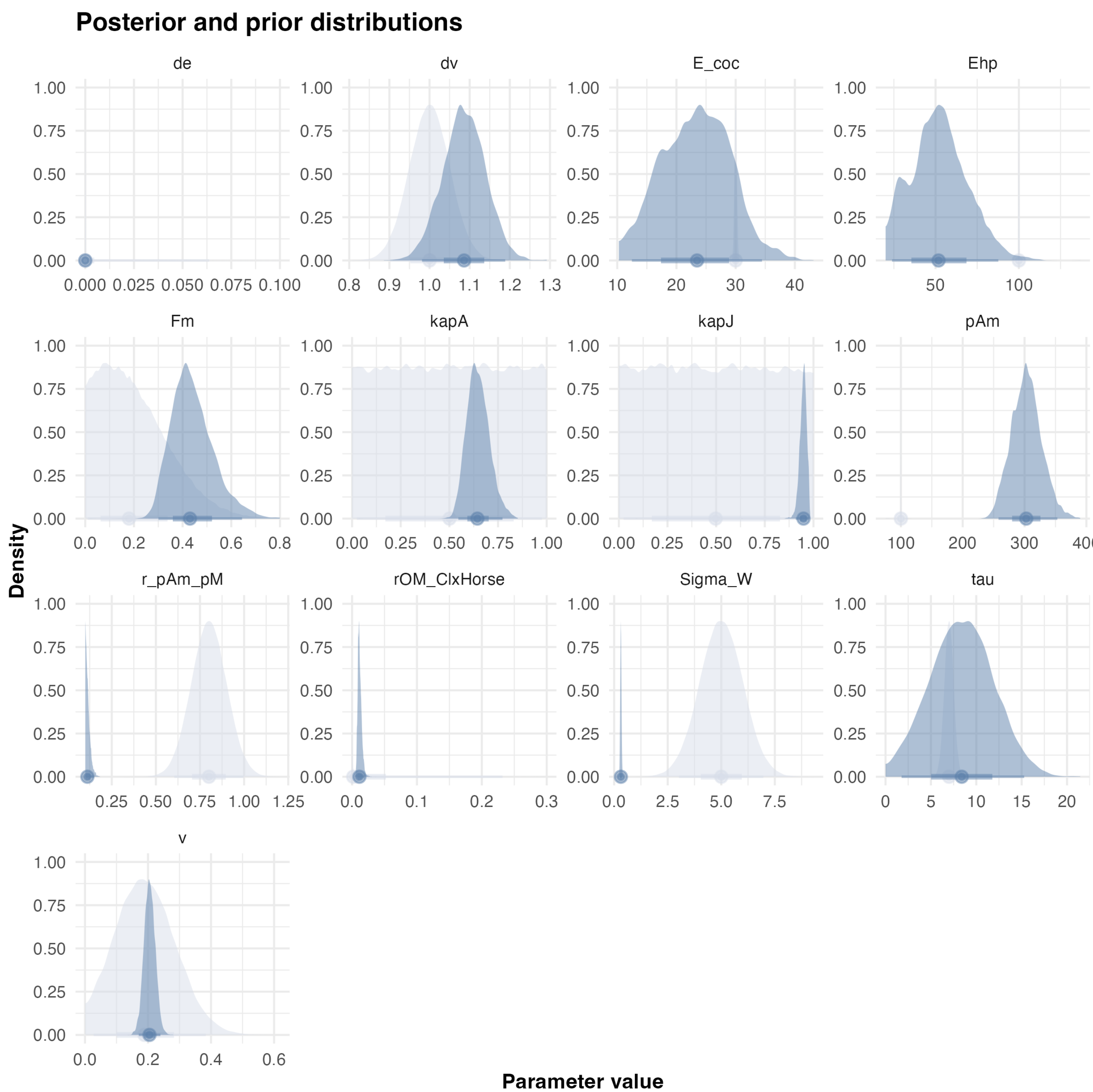


Figure 3.6: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

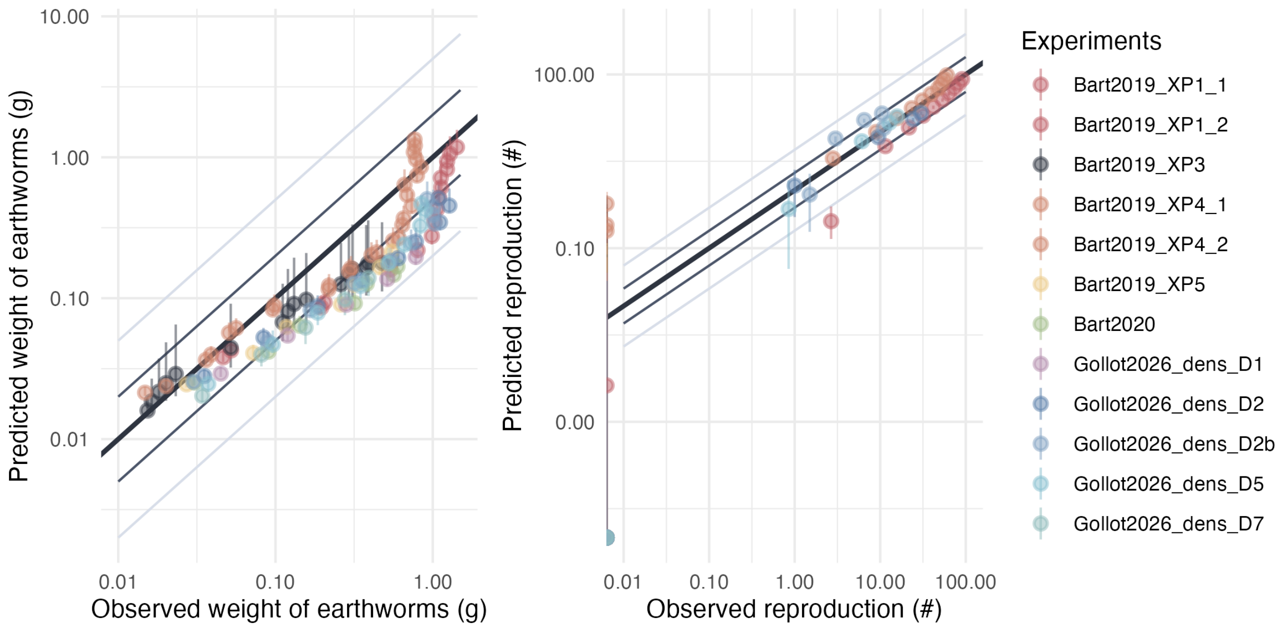


Figure 3.7: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

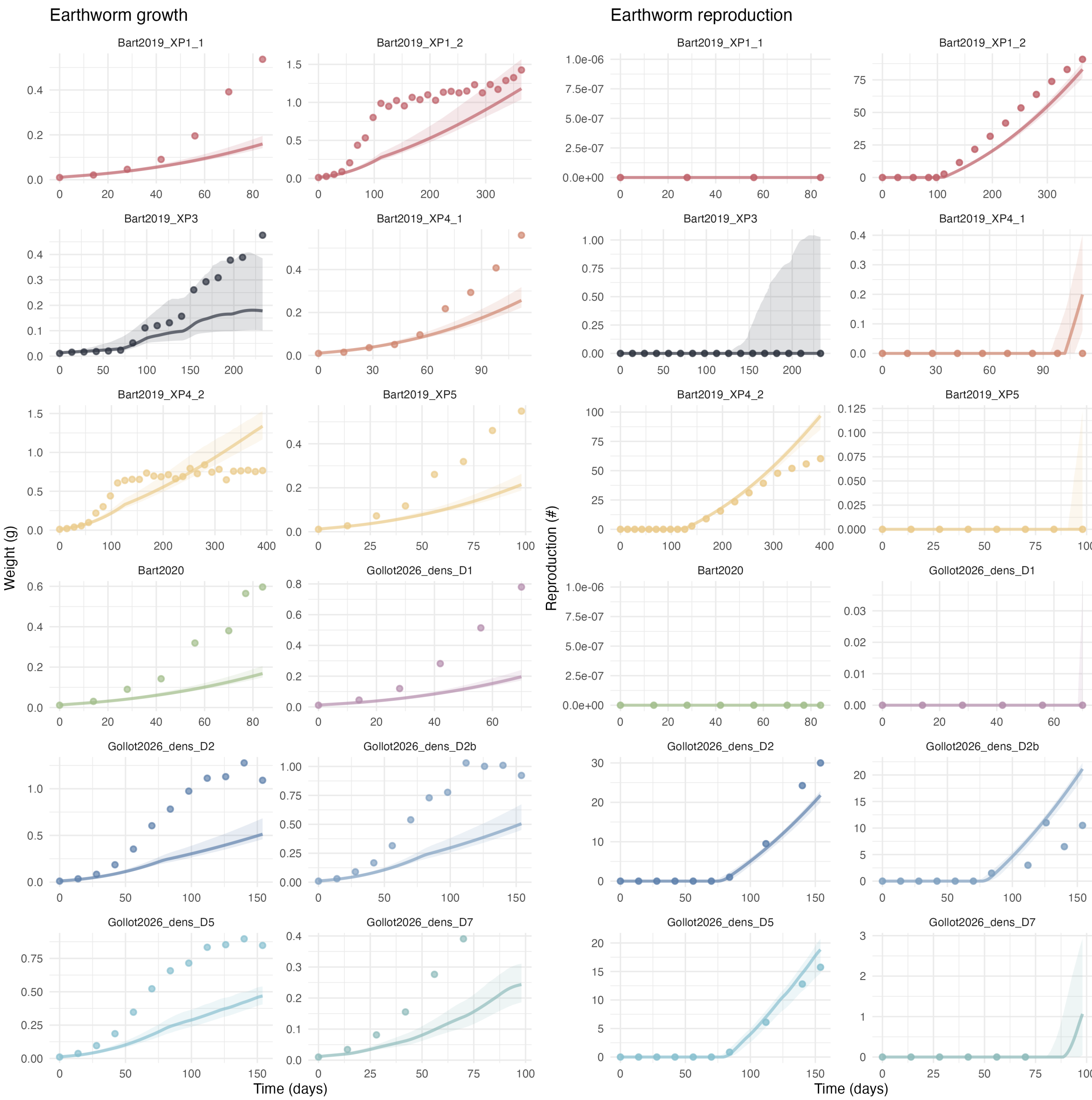


Figure 3.8: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.6 Model D_CWKe

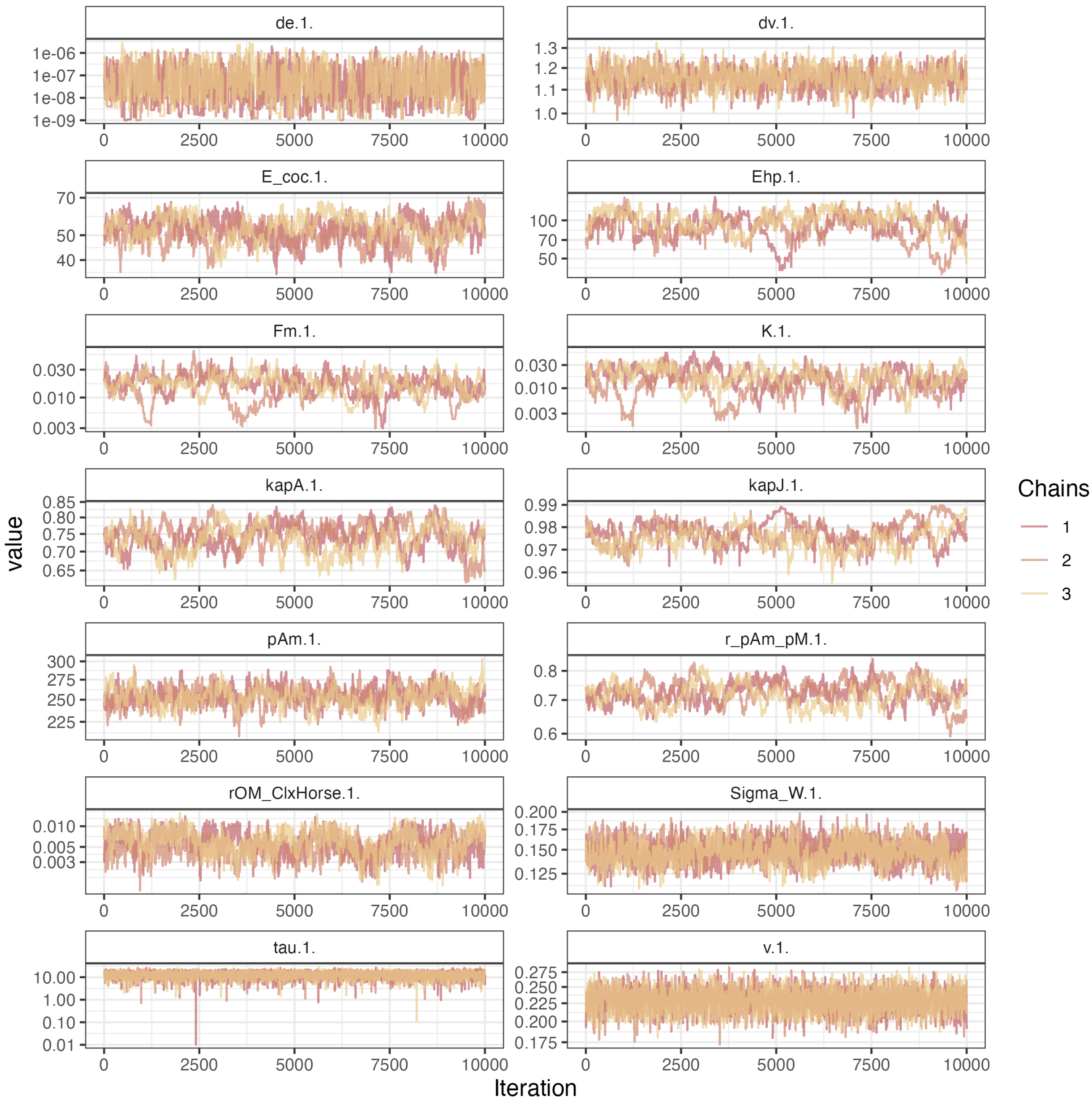


Figure 3.9: Model chains.

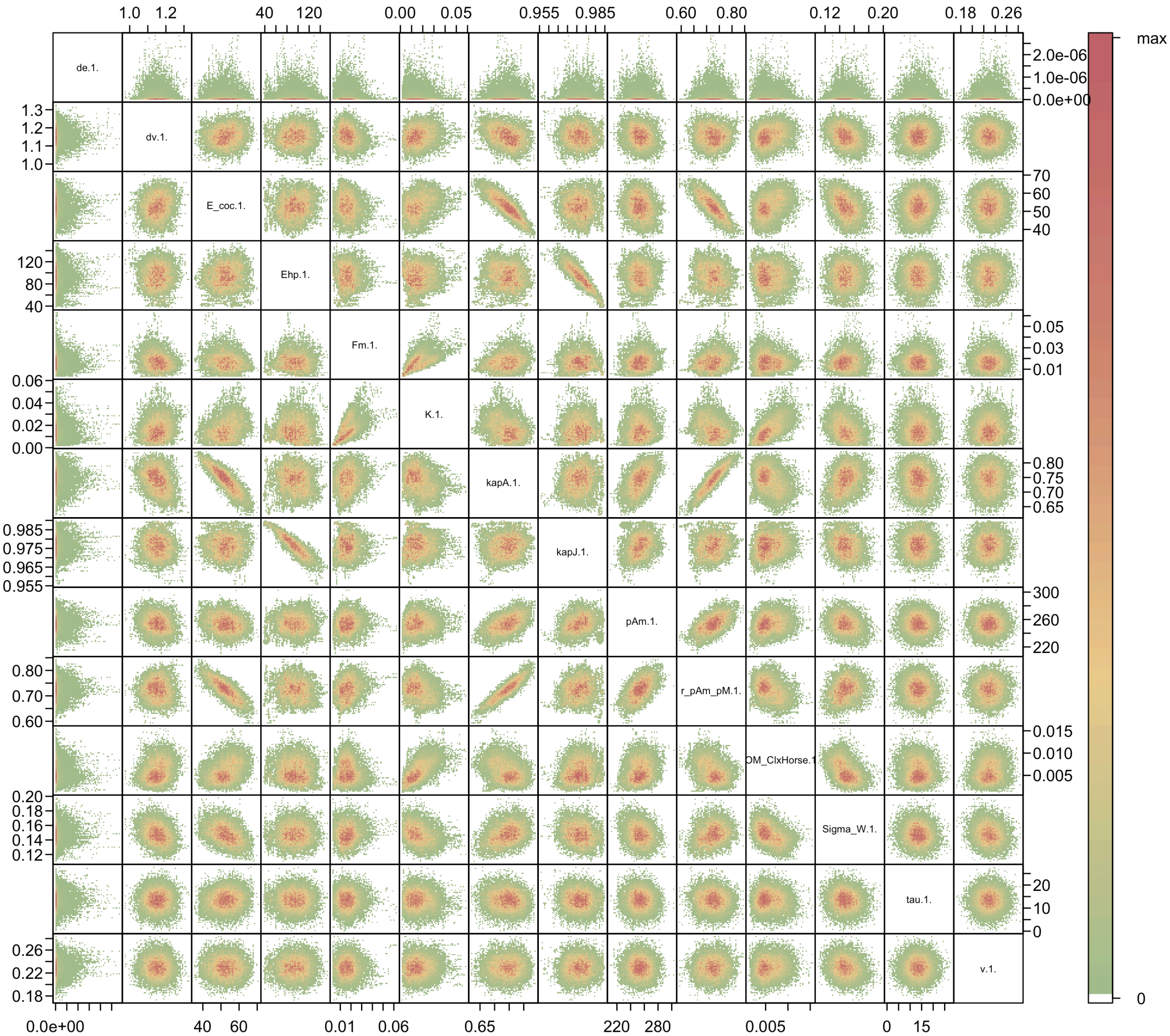


Figure 3.10: Chain correlations.

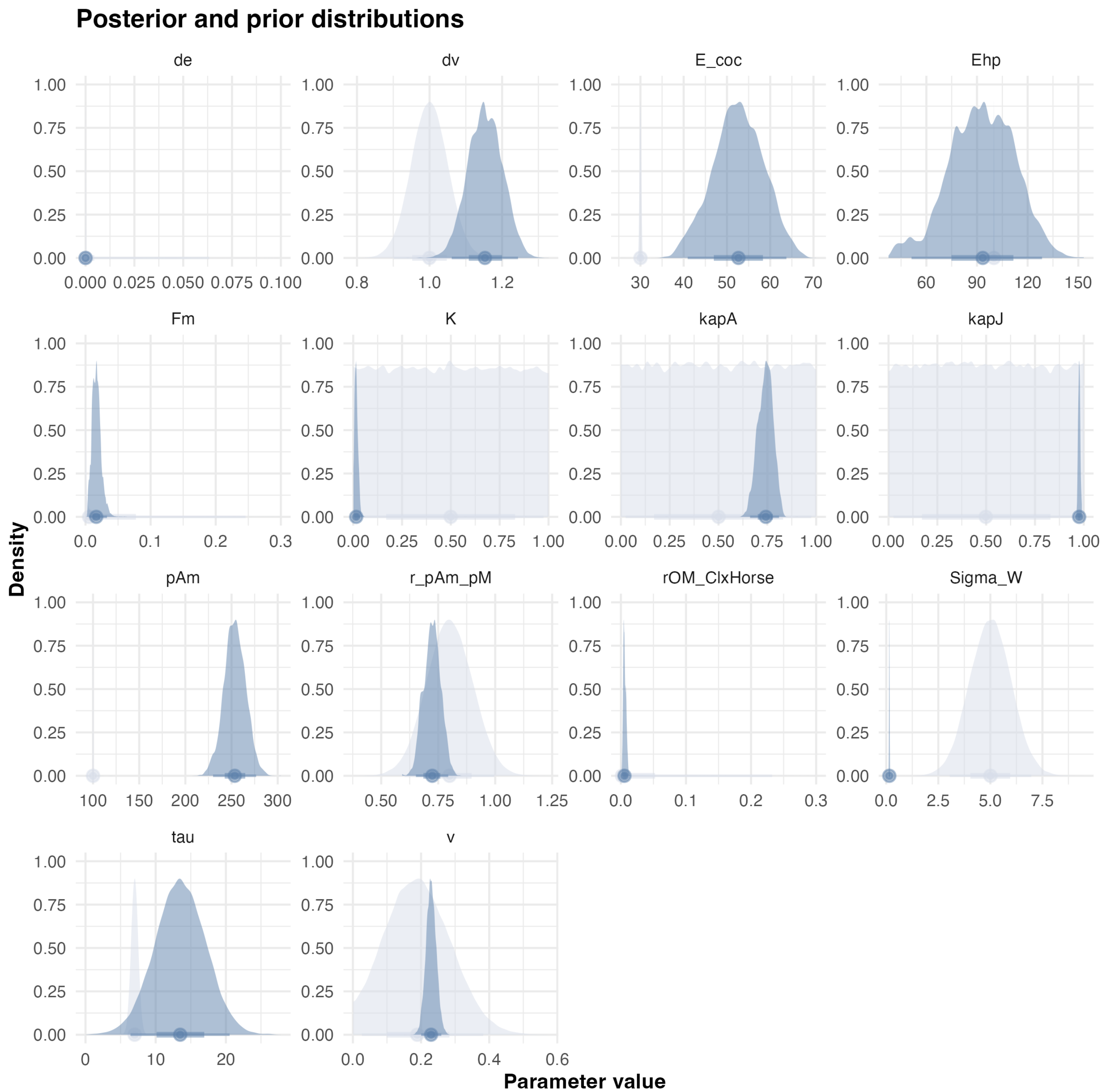


Figure 3.11: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

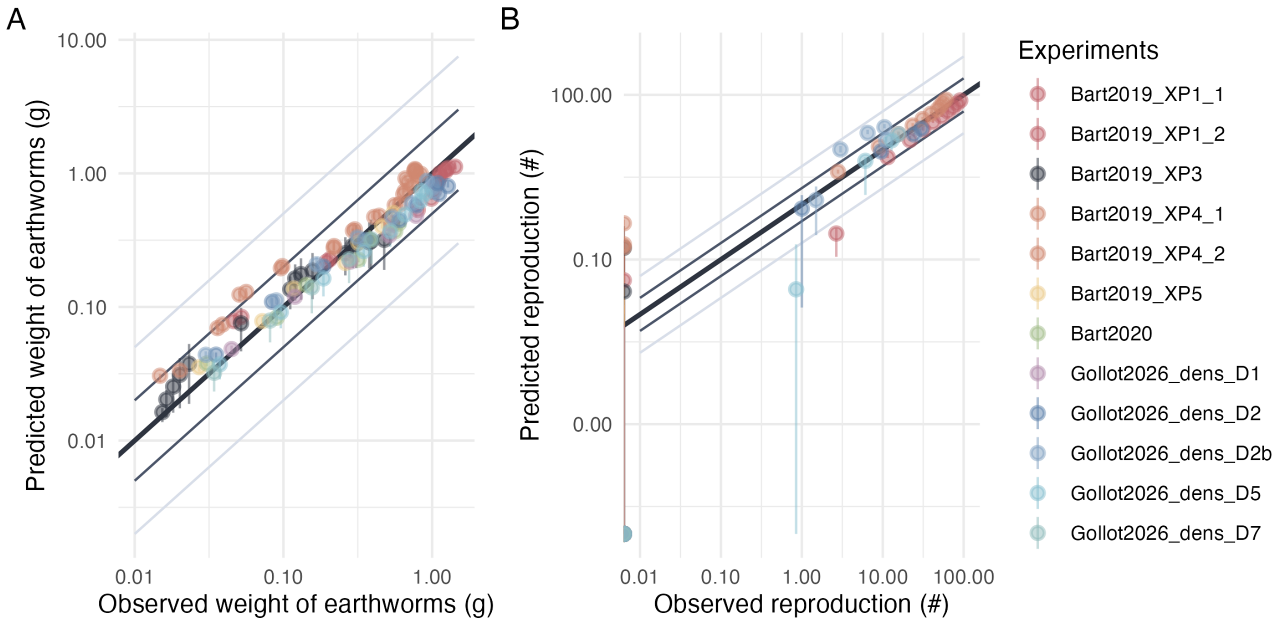


Figure 3.12: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

3.7 Model E_CLYe

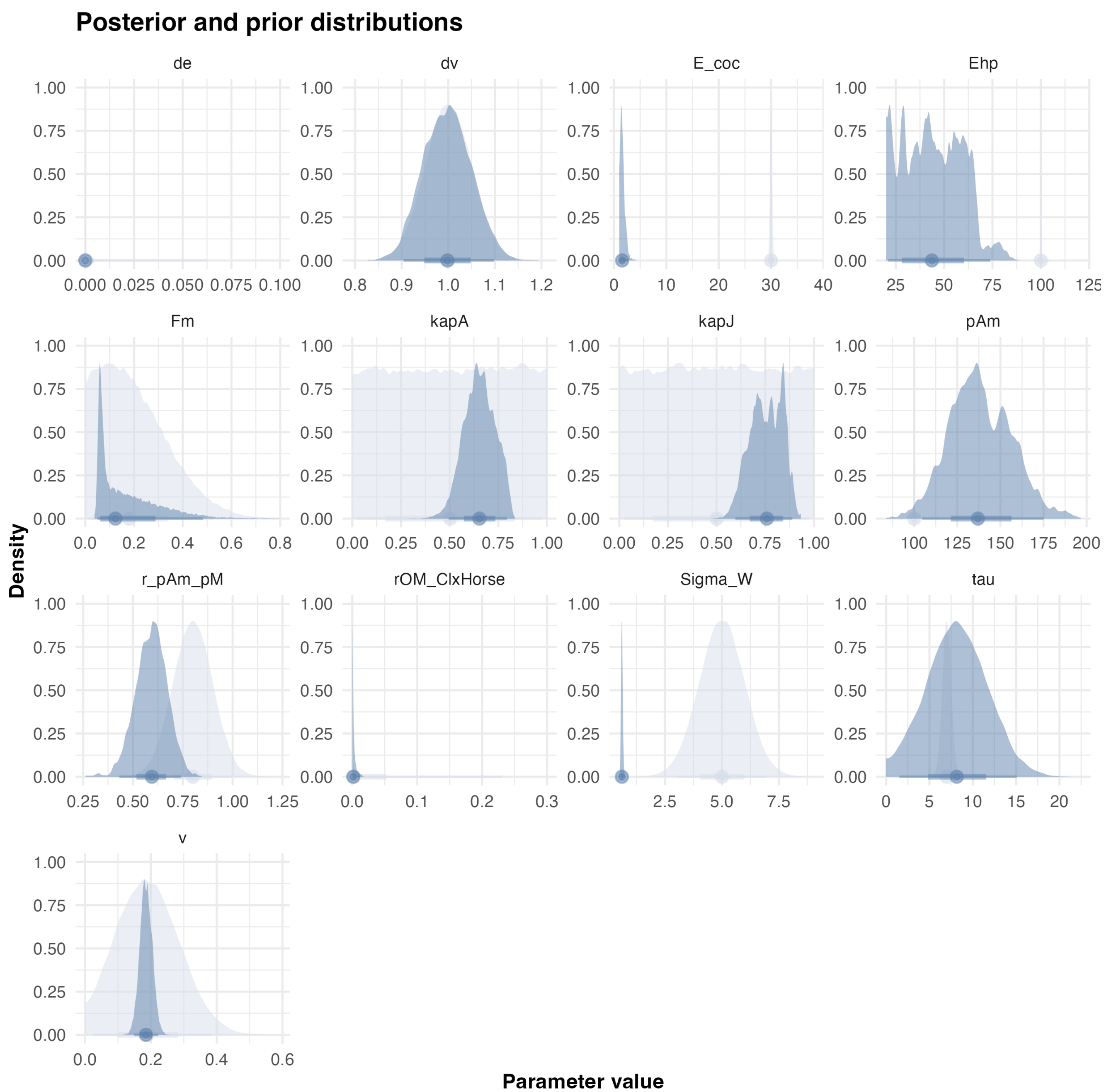


Figure 3.13: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

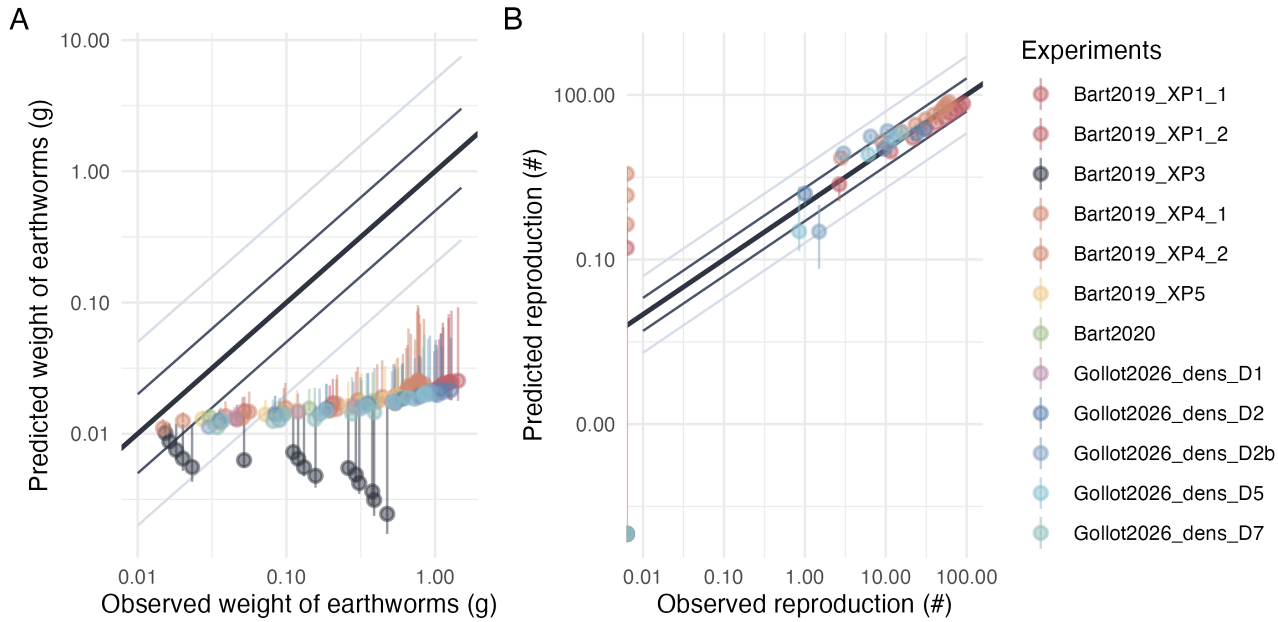


Figure 3.14: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

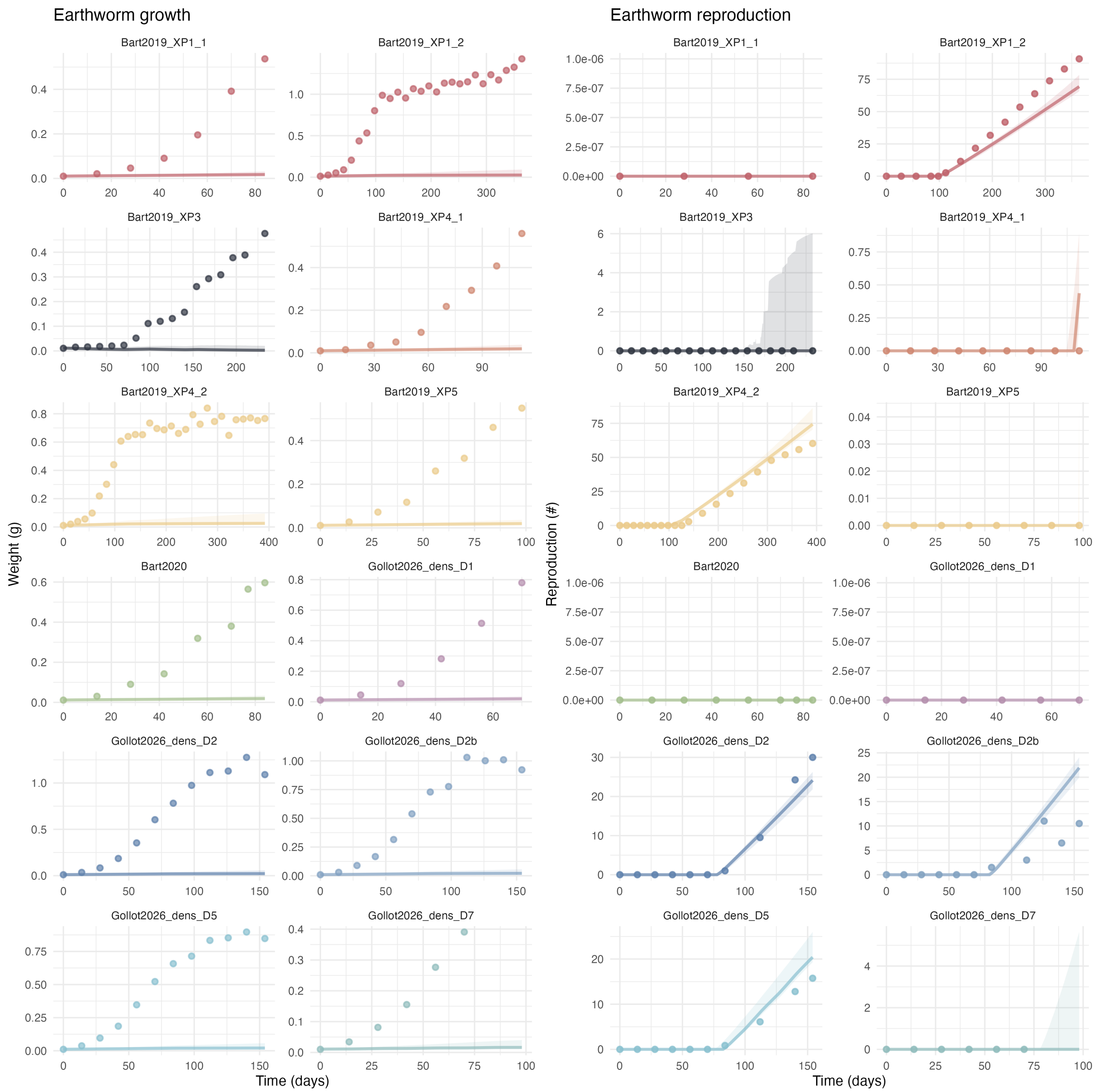


Figure 3.15: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.8 Model F_CLKe

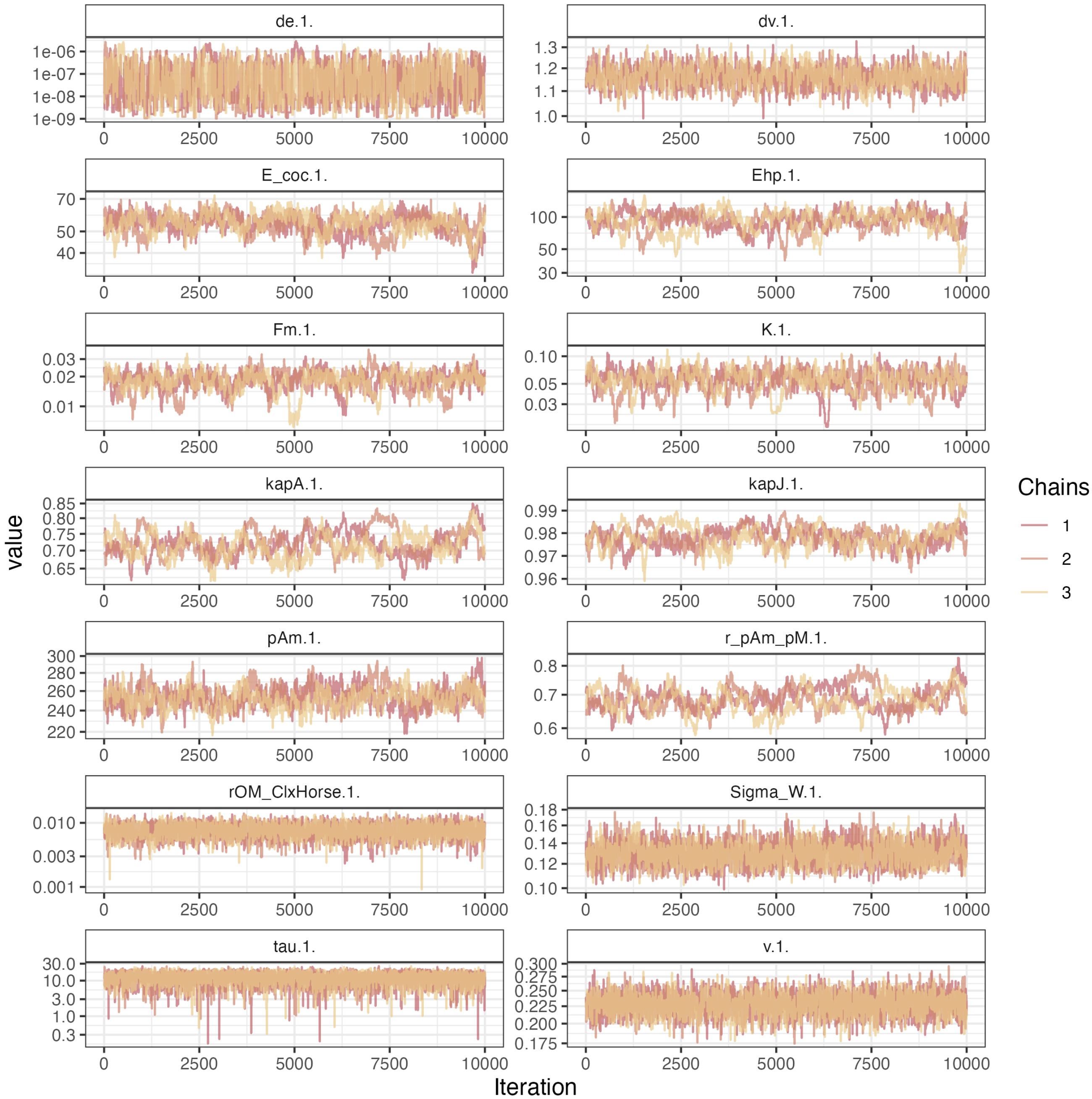


Figure 3.16: Model chains.

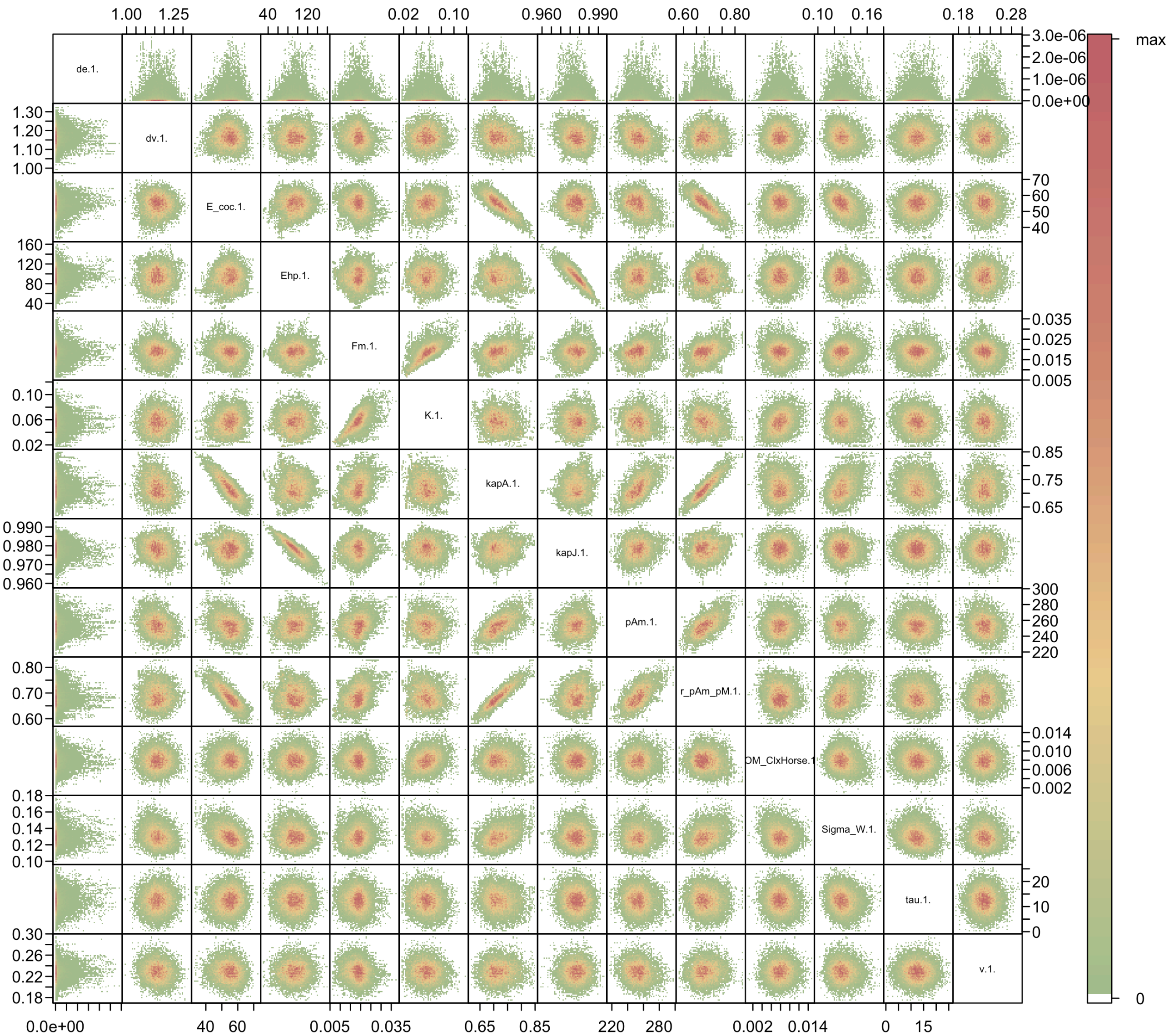


Figure 3.17: Chain correlations.

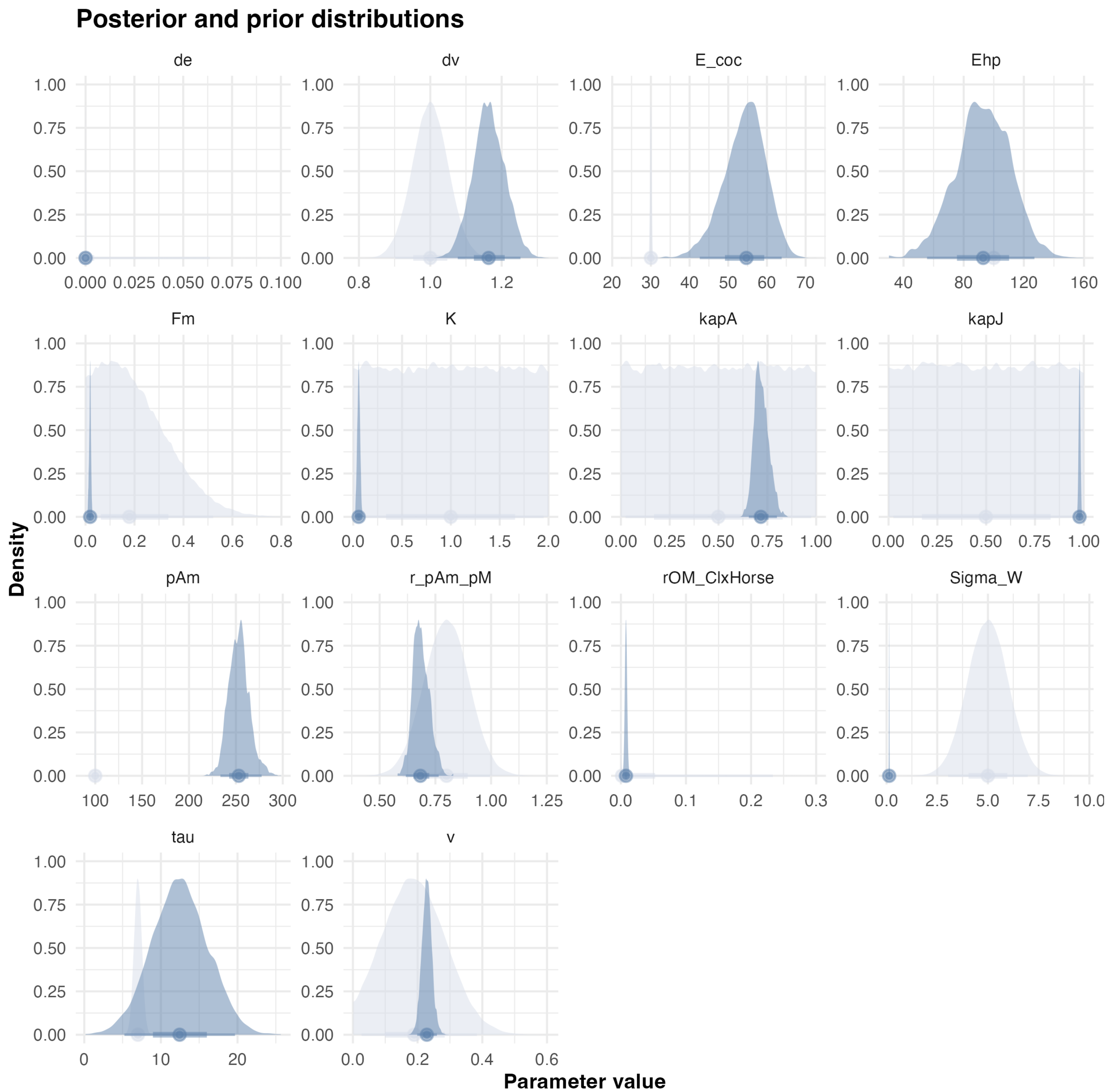


Figure 3.18: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

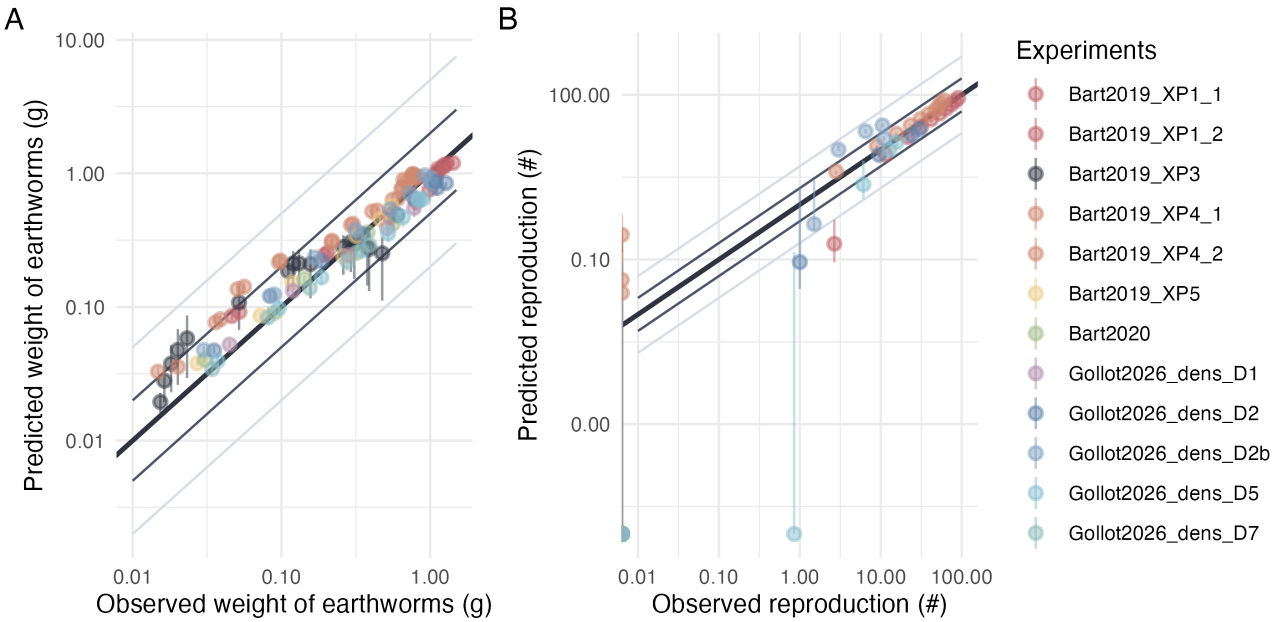


Figure 3.19: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

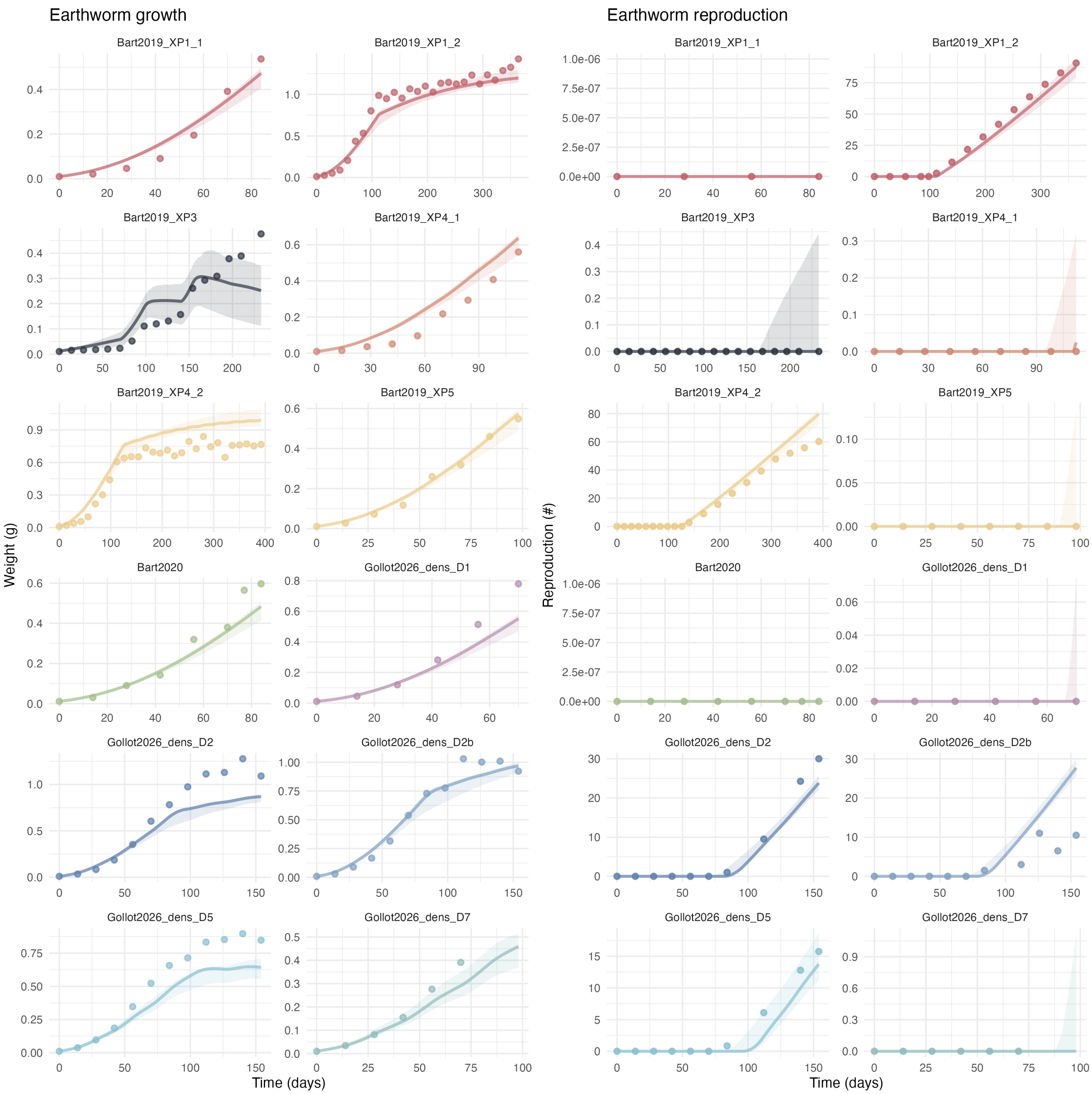


Figure 3.20: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.9 Summary

Table 3.2: Estimate summary of each models and their respective log-likelihood and BIC.

Parameter	A_FLYw	B_FLYe	C_CWYe	D_CWKe	E_CLYe	F_CLKe
$\{\dot{p}_{Am}\}$	256.6	281.6	291.3	254.5	111.1	254.7
	[240.6 ; 276.3]	[230.0 ; 316.8]	[258.0 ; 353.0]	[230.0 ; 276.6]	[104.7 ; 175.1]	[233.5 ; 277.4]
$[\dot{p}_M]/\{\dot{p}_{Am}\}$ ratio	1.09	0.17	0.76	0.755	0.67	0.68
	[0.99 ; 1.13]	[0.12 ; 0.19]	[0.65 ; 0.79]	[0.617 ; 0.896]	[0.43 ; 0.74]	[0.62 ; 0.76]
$\{\dot{F}_m\}$ or $[\dot{F}_m]$	0.17	0.13	0.38	0.019	0.05	0.019
	[0.12 ; 0.47]	[0.11 ; 0.28]	[0.30 ; 0.64]	[0.005 ; 0.033]	[0.05 ; 0.48]	[0.010 ; 0.026]
r_{OM}	0.011	0.004	0.01	0.005	0.0004	0.007
	[0.004 ; 0.015]	[0.004 ; 0.012]	[0.007 ; 0.019]	[0.002 ; 0.011]	[0.0003 ; 0.012]	[0.004 ; 0.011]
K	NA	NA	NA	0.014	NA	0.05
				[0.003 ; 0.035]		[0.03 ; 0.08]
v	0.20	0.19	0.21	0.22	0.18	0.23
	[0.16 ; 0.23]	[0.16 ; 0.23]	[0.17 ; 0.24]	[0.20 ; 0.26]	[0.15 ; 0.22]	[0.20 ; 0.26]
κ	0.63	0.92	NA	NA	NA	NA
	[0.52 ; 0.74]	[0.86 ; 0.94]				
κ_J	NA	NA	0.94	0.98	0.74	0.98
			[0.91 ; 0.98]	[0.97 ; 0.99]	[0.60 ; 0.957]	[0.97 ; 0.99]
κ_A	NA	NA	0.60	0.77	0.61	0.71
			[0.55 ; 0.77]	[0.66 ; 0.81]	[0.49 ; 0.79]	[0.65 ; 0.80]
τ	NA	NA	7.1	13.3	6.8	14.4
			[1.8 ; 15.3]	[6.4 ; 20.5]	[1.5 ; 15.0]	[5.3 ; 19.7]
E_h^p	24.5	64.9	52.1	78.8	36.5	103.5
	[21.2 ; 29.0]	[41.5 ; 105.4]	[23.9 ; 87.7]	[51.2 ; 128.5]	[21.0 ; 73.8]	[55.6 ; 127.0]
E_{coc}	1.0	4.7	24.0	46.9	1.2	56.2
	[1.0 ; 1.2]	[2.8 ; 7.6]	[12.5 ; 34.4]	[40.9 ; 63.7]	[1.0 ; 2.7]	[42.6 ; 63.8]

Parameter	A_FLYw	B_FLYe	C_CWYe	D_CWKe	E_CLYe	F_CLKe
d_V	1.0 [0.9 ; 1.1]	1.1 [0.94 ; 1.2]	1.1 [1.0 ; 1.2]	1.1 [1.1 ; 1.2]	1.0 [0.9 ; 1.1]	1.2 [1.1 ; 1.3]
d_E	NA	1.1e-9 [1.1e-9 ; 5.3e-7]	1.0e-9 [1.2e-9 ; 3.4e-7]	1.7e-9 [1.2e-9 ; 8.4e-6]	1.0e-9 [1.4e-9 ; 7.9e-6]	1.6e-9 [1.2e-9 ; 1.1e-6]
w_E	467.9 [249.3 ; 735.1]	NA	NA	NA	NA	NA
σ_W	0.16 [0.14 ; 0.19]	0.35 [0.31 ; 0.41]	0.32 [0.26 ; 0.37]	0.15 [0.12 ; 0.17]	0.64 [0.54 ; 0.69]	0.12 [0.11 ; 0.15]
Log-likelihood	-77.5	-199.1	-171.8	-55.8	-266.1	-38.0
BIC	211.2	454.5	411.1	184.7	599.7	149.2

Figure 3.21: RMSE for Weight and Reproduction per experiment for each tested model.

4 Isolated adults

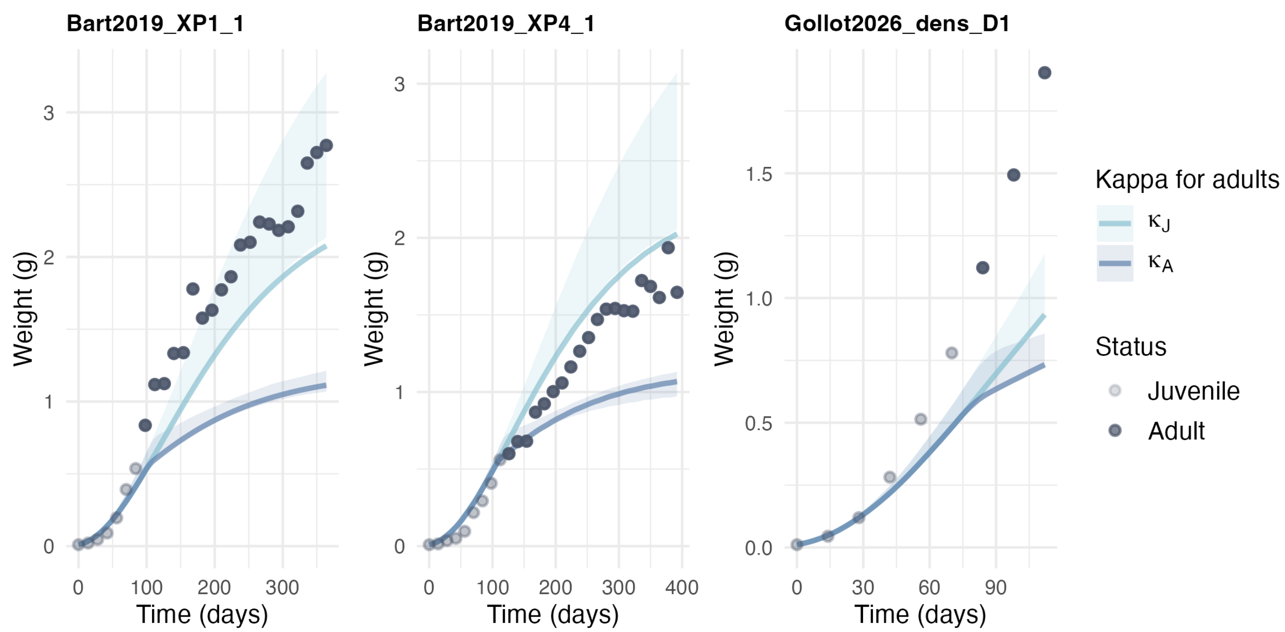


Figure 4.1: Observed and predicted growth trajectories for isolated adults simulated with the retained model (CWKe) under formulations with either a constant juvenile allocation fraction (κ_J, light blue lines) or a transition toward an adult allocation fraction (κ_A, dark blue lines). Solid lines correspond to estimated model trajectories with transparent ribbons corresponding to the 95% credibilitty intervals. Transparent points represent observations from the juvenile stage, whereas filled points represent observations from the adult stage.

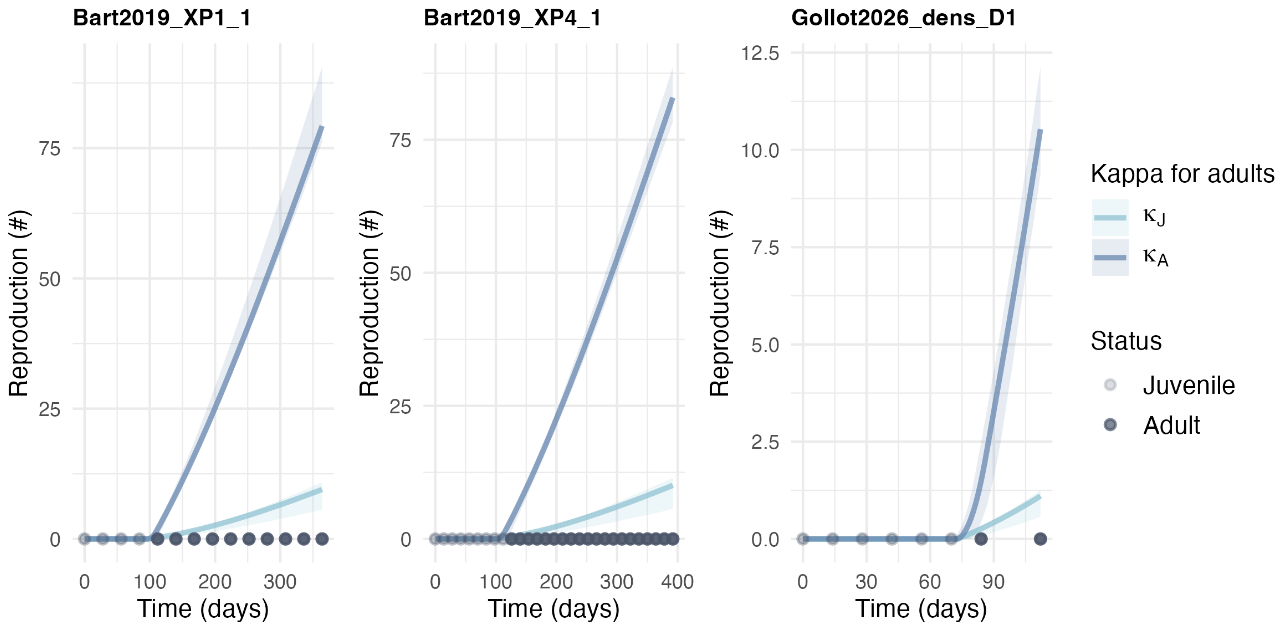


Figure 4.2: Observed and predicted reproduction trajectories for isolated adults simulated with the retained model (CWKe) under formulations with either a constant juvenile allocation fraction (κ_J, light blue lines) or a transition toward an adult allocation fraction (κ_A, dark blue lines). Solid lines correspond to estimated model trajectories with transparent ribbons corresponding to the 95% credibilitty intervals. Transparent points represent observations from the juvenile stage, whereas filled points represent observations from the adult stage.

Bart, S. (2019). Towards a better understanding of the life cycle of the earthworm Aporrectodea caliginosa: New data and energy-based modelling. *Pedobiologia*, *77*, 150592. https://doi.org/10.1016/j.pedobi.2019.150592

Bart, S., Laurent, C., Péry, A. R. R., Mougin, C., & Pelosi, C. (2017). Differences in sensitivity between earthworms and enchytraeids exposed to two commercial fungicides. *Ecotoxicology and Environmental Safety*, *140*, 177–184. https://doi.org/10.1016/j.ecoenv.2017.02.052

Bart, S., Pelosi, C., Nélieu, S., Lamy, I., & Péry, A. R. R. (2020). An energy-based model to analyze growth data of earthworms exposed to two fungicides. *Environmental Science and Pollution Research*, *27*(1), 741–750. https://doi.org/10.1007/s11356-019-06985-z

Cumminns, K. W., & Wuycheck, J. C. (1971). Caloric Equivalents for Investigations in Ecological Energetics. *SIL Communications, 1953-1996*, *18*(1), 1–158. https://doi.org/10.1080/05384680.1971.11903918

Driver, E. A., Sugden, L. G., & Kovach, R. J. (1974). Calorific, chemical and physical values of potential duck foods. *Freshwater Biology*, *4*(3), 281–292. https://doi.org/10.1111/j.1365-2427.1974.tb00098.x

Eggleton, M. A., & Schramm, H. L. (2004). Feeding Ecology and Energetic Relationships with Habitat of Blue Catfish, Ictalurus furcatus, and Flathead Catfish, Pylodictis olivaris, in the Lower Mississippi River, U.S.A. *Environmental Biology of Fishes*, *70*(2), 107–121. https://doi.org/10.1023/B:EBFI.0000029341.45030.94

Kabus, J., Hartmann, V., Cocchiararo, B., Dombrowski, A., Enns, D., Karaouzas, I., Lipkowski, K., Pelikan, L., Shumka, S., Soose, L., Baker, N. J., & Jourdan, J. (2024). Cryptic species complex shows population-dependent, rather than lineage-dependent tolerance to a neonicotinoid. *Environmental Pollution*, *362*, 124888. https://doi.org/10.1016/j.envpol.2024.124888